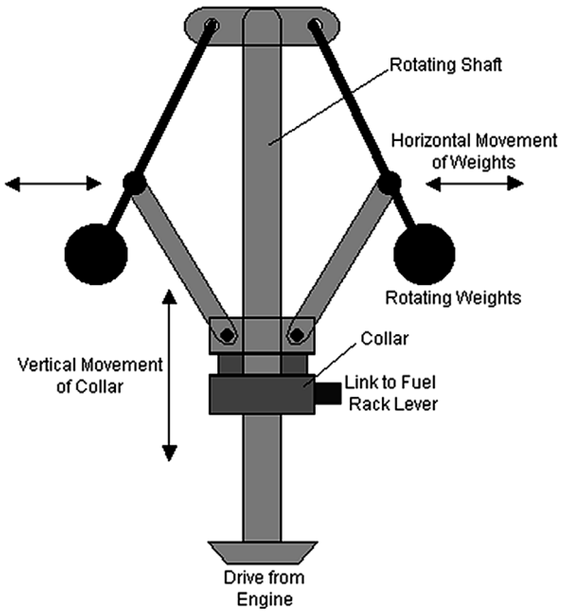


Fig 9.10.1.1 Layout of a Mechanical Gover**nor**

Working Principle: When the engine is started the governor shaft gets the power and rotates. As the governor shaft rotates the masses connected with the spring also rotate and hence the masses are forced to rotate outside due to the centrifugal force. Hence the sleeve placed in the governor shaft is moved upwards. The one end of the lever connected with the control rack is moved up along with the sleeve. This causes the other end of the lever to move down and operates the control rack. Due to this the diesel supplied to the engine is reduced and hence the engine speed is reduced. Similarly when the accelerator pedal of the driver is pressed the control rack is operated and the speed of the engine is varied. In the same way the engine speed can also be increased by moving the centrifugal mass. The governors of this type are of two types. 1. All speed governor and 2. Maximum speed governor. The governor operated by the driver's engine speed control is called as the maximum speed governor and the governor operated by the spring force without the accelerator is called as all speed governor.

2. **Pneumatic Governor :** The governor operated by using only the engine Vacuum without any bolts is called as the pneumatic governor.

Construction: This type of governor has two main parts such as 1. Venturi unit and 2. Diaphragm unit. The venturi unit is connected with the intake manifold and the diaphragm unit is connected with the fuel pump. The venturi unit and diaphragm unit are connected by the Vacuum pipe. The diaphragm is connected to the control rack. The construction of the pneumatic governor is shown in Fig 9.10.1.2.

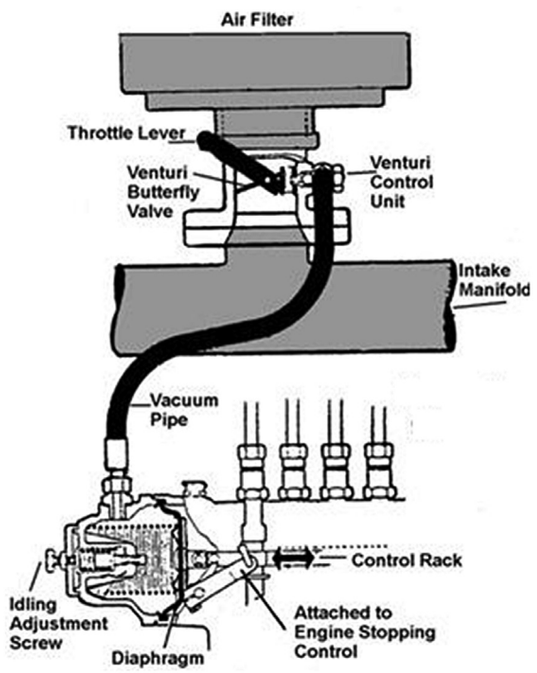


Fig 9.10.1.2 Pneumatic Governor

Working Principle: When the accelerator pedal is pressed the butterfly valve in the venturi opens. Hence the Vacuum in the inlet manifold reaches the diaphragm unit

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